

# Lower bounds for nontrivial cycles of the Juggler map

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## Abstract

The Juggler map is the nonlinear integer map

$$J(n) = \begin{cases} \lfloor \sqrt{n} \rfloor, & n \text{ even,} \\ \lfloor n^{3/2} \rfloor, & n \text{ odd.} \end{cases}$$

It is conjectured that every positive integer eventually reaches 1. This note does not prove that conjecture, and it does not exclude every cycle. It restricts the possible periods of a hypothetical nontrivial cycle.

A realized parity word obeys a power envelope, so a cycle word is formally expanding. Floor remainders must finance the surplus  $3^o - 2^L$ . At a cycle minimum this yields

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

Combined with the computational verification reported by Weisstein that every start through  $10^6$  reaches 1, there is no nontrivial cycle of length at most 1053. Through length  $10^5$ , only an explicit set  $\mathcal{E}$  of 397 lengths remains admissible to the bound. A separate finite-word analysis shows that every nontrivial cycle has at least four even letters and hence period at least eleven.

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## 1. Introduction

The Juggler sequence was introduced by Pickover [1,2]. The one-step map is OEIS A094683 [3]; the stopping-time table is A007320 [4]. The orbit of 3 is 3, 5, 11, 36, 6, 2, 1. The orbit of 37 already peaks at 24906114455136. Universal arrival at 1 remains the open conjecture of the subject.

The map combines a contracting even branch with an expanding odd branch,

$$E(n) = \lfloor n^{1/2} \rfloor, \quad O(n) = \lfloor n^{3/2} \rfloor.$$

A word of length  $k$  with  $o$  odd letters has ideal exponent  $3^o/2^k$ . Floors are applied after every letter, and a word is available only when the orbit realizes those parities.

The note has one architecture:

local word obstructions + global finance  $\Rightarrow$  a cycle-length lower bound.

Section 3 excludes every cycle word with fewer than four even letters, so the period is at least eleven. Section 4 turns the formal surplus  $3^o - 2^L$  into a bound on the cycle minimum. With the verified descent floor through  $10^6$ , that bound excludes every period at most 1053.

Write  $N_0 = 10^6$  for the *verified descent floor*: the computational verification reported by Weisstein [5], recomputed here by exact first-passage. It is not a new computational record. The contribution is the implication

$$\text{finance} + \text{verified descent floor } 10^6 \Rightarrow L \geq 1054.$$

Throughout,  $\mathbb{N} = \{1, 2, 3, \dots\}$ . Write  $J^k$  for the  $k$ -fold iterate. A nonempty realized word  $w$  with  $J^{|w|}(n) = n$  is a *cycle word*. The unique fixed point is 1; a cycle is *nontrivial* when it contains some  $n \geq 2$ . A cycle word at  $n$  is *minimum-based* when  $n$  is a cycle minimum:  $J^j(n) \geq n$  for every  $0 \leq j < |w|$ . Every cycle word has a minimum-based rotation.

## 1.0 Main results

**Main theorem.** There is no nontrivial Juggler cycle of length at most 1053. Equivalently, any nontrivial cycle has period at least 1054 (Theorem A; Theorem 4.6(A)).

**Secondary theorem.** If a nontrivial cycle has period  $L \leq 10^5$ , then  $L$  belongs to an explicit set  $\mathcal{E}$  of 397 lengths (Theorem B; Theorem 4.6(B)). Membership in  $\mathcal{E}$  means only that the finance inequality at the verified descent floor  $10^6$  does not exclude  $L$ . It does not mean that a cycle of that length exists, or that the length is dynamically plausible.

**Supporting structural theorem.** Every nontrivial cycle word has at least four even letters, and hence period at least eleven (Theorem C; Theorem 3.22 and Corollary 3.23). This bound does not use the verified descent floor.

Theorems A and B are the finance inequality of Theorem 4.4, in the conservative 6/5 form of Corollary 4.5, at the verified descent floor  $N_0 = 10^6$  reported by Weisstein [5]. These results give no control over the remaining lengths in  $\mathcal{E}$ , and they do not imply termination.

The logical dependency is

word envelope  $\rightarrow$  cycle minimum  $\rightarrow$  finance  $\rightarrow n_{\max}(L) \rightarrow 10^6$  verified descent floor  $\rightarrow L \geq 1054$ .

## 1.1 Related work

Pickover’s later exposition is Chapter 45 of [2]. Weisstein [5] records the map, the stopping-time sequences A094670, A094679, A095908, and a verification of arrival at 1 through  $10^6$ . That verified descent floor is the input to Theorem 4.6; the first-passage run of Appendix B recomputes it and is not a new record. OEIS A094716 [6] records extreme heights, including the start 48443 whose peak has 972 463 digits. Those height records do not bound the period.

We found no published result establishing an explicit lower bound on the period of a nontrivial cycle for this exact floor-power Juggler map. Prasad–Prasad [7] estimate excursion and stopping constants for juggler-like maps by a random-walk large-deviation model; those estimates do not apply to exact cycles. Small-cycle censuses are a standard first layer for Collatz-like maps, surveyed by Lagarias [8,9]. Those results do not transfer: the branches of  $J$  are floor powers rather than affine maps (Crandall [10], Matthews–Watts [11]). In particular there is no identity of the form  $n(2^K - 3^p) = C$ .

The financing-versus-gap template of Section 4 follows Simons and de Weger [12] on Collatz  $m$ -cycles. That paper uses Diophantine approximation and linear forms in logarithms. Financing as an idea, logarithmic step inequalities, and the continued-fraction phenomenon are not claimed as new. The argument below is elementary and independent of those Diophantine tools: floor-power defects are relatively  $O(1/x)$  in logarithms, so a uniform logarithmic floor-error bound, valid above the verified floor, excludes every length that is not admissible for Theorem 4.4. The Juggler-specific form of the financing inequality, and the explicit period bounds it produces with the verified descent floor  $10^6$ , appear to be new. The leftover packaging, period at least  $X$  or a named admissible family, follows Eliahou [13]. For this nonlinear floor-power map, the elementary log-envelope plus that floor already forces  $L \geq 1054$ .

**Novelty statement.** For the exact Juggler map, the one-step floor cells yield the cycle-minimum inequality

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o,$$

and hence a computable function  $n_{\max}(L)$  giving explicit period exclusions (Theorem 4.4 and Corollary 4.5). To our knowledge, this explicit cycle-period bound has not previously been derived for the exact Juggler map.

## 1.2 Verification

The arguments of Sections 2, 3, and 4 may be read without machine assistance. A Lean formalization of those arguments is recorded in Appendix A. The finite tables used by Lemmas 3.5, 3.7, 3.11 and Theorems 3.12–3.20 are `native_decide` evaluations in the modules named there. They are finite computations, not a termination proof.

Theorem 4.6 uses an exact-integer first-passage run through  $10^6$ , together with

an exact gap table of lengths up to  $10^5$ . The floor itself is the verified descent floor reported by Weissstein [5]; the first-passage run is an independent recomputation of that same floor, not a new record. Checksums and regeneration commands are in Appendix B.

```
Repository: https://github.com/sneakyweasel/balanced_ternary
Commit:    7802f78bec58c68cec92a2efb3db4a502f916277
Lean:      leanprover/lean4:v4.33.0
Mathlib:   v4.33.0 (lake-manifest rev db584cd6d46c92f209a44c0f1c829460d327499d)
Build:      lake build Problems.JugglerPaper (from formal/)
Computation: python -m research.juggler_sequence.cycle_finance
SHA-256:    Appendix B
```

The commit is the repository state that produced the finance tables. A later editorial commit of this text does not change those hashes.

## 2. Envelope and defect

Let  $\mathcal{B} = \{E, O\}$ . A finite word  $w \in \mathcal{B}^*$  is *realized* at  $n \in \mathbb{N}$  when the successive parities of the orbit of  $n$  are exactly the letters of  $w$ . Write  $J^{|w|}(n)$  for the endpoint after those letters, and  $\#O(w)$  for the number of odd letters. When  $w$  is realized, this endpoint is the  $|w|$ -fold iterate.

**Theorem 2.1 (fixed-word monotonicity).** If  $n \leq m$  and both realize  $w$ , then  $J^{|w|}(n) \leq J^{|w|}(m)$ .

*Proof.* Induct on  $w$ . The empty word is immediate. The images after a common realized prefix remain ordered, and both current states realize the same next letter. The even branch is the monotone integer square root; the odd branch is  $x \mapsto x^3$  followed by the integer square root.  $\square$

The realizing set of a fixed word need not be an interval:  $OE$  is realized at 7 and at 11 but not at 9.

**Theorem 2.2 (finite-word power envelope).** If  $w$  is realized at  $n$  and  $m = J^{|w|}(n)$ , then

$$m^{2^{|w|}} \leq n^{3^{\#O(w)}}.$$

*Proof.* The empty word is the equality  $n \leq n$ . Suppose a realized prefix of length  $\ell$  with odd count  $o$  ends at  $x$  and satisfies  $x^{2^\ell} \leq n^{3^o}$ , and the next letter is realized.

If the next letter is even, then  $J(x)^2 \leq x$ . Raising the inductive bound to the second power gives

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{2^\ell} \leq n^{3^o}.$$

The odd count is unchanged.

If the next letter is odd, then  $J(x)^2 \leq x^3$ , so

$$J(x)^{2^{\ell+1}} = (J(x)^2)^{2^\ell} \leq x^{3 \cdot 2^\ell} = (x^{2^\ell})^3 \leq (n^{3^o})^3 = n^{3^{o+1}}.$$

This is the claimed bound after one more odd letter.  $\square$

**Corollary 2.3 (exponent-gap contraction).** If  $n \geq 2$ ,  $w$  is realized at  $n$ , and  $3^{\#O(w)} < 2^{|w|}$ , then  $J^{|w|}(n) < n$ .

*Proof.* Let  $m = J^{|w|}(n)$  and  $k = |w|$ . Theorem 2.2 gives  $m^{2^k} \leq n^{3^{\#O(w)}}$ . The exponent gap and  $n \geq 2$  give  $n^{3^{\#O(w)}} < n^{2^k}$ , so  $m^{2^k} < n^{2^k}$ . Since  $m \geq 1$ , one has  $m < n$ .  $\square$

The corollary includes familiar contracting blocks such as  $OOOEE$ . It does not prove that every start realizes some contracting word.

## 2.4 Exact floor defect

Section 4 uses only the envelope (Theorem 2.2), the one-step cells, and  $\log(1+u) \leq u$ . The following records the exact multiplicative identity behind the envelope and its vanishing cases. The composition law is Theorem 2.6 in Appendix C (`global_defect_append`); it is not an input to Theorem 4.4.

For a single branch,

$$x^e = J(x)^2 + \rho(x), \quad e = \begin{cases} 1, & x \text{ even,} \\ 3, & x \text{ odd,} \end{cases}$$

with  $0 \leq \rho(x) < 2J(x) + 1$ . Write  $\text{gap}(a, \rho, e) = (a + \rho)^e - a^e$ . The *global defect*  $\Delta_w(n)$  is the terminal value of the resulting power-gap recurrence.

**Theorem 2.4 (global defect identity).** If  $w$  is realized at  $n$  and  $m = J^{|w|}(n)$ , then

$$n^{3^{\#O(w)}} = m^{2^{|w|}} + \Delta_w(n), \quad \Delta_w(n) \geq 0.$$

Theorem 2.2 is the inequality  $\Delta_w(n) \geq 0$ .

*Proof.* Induct on  $w$ . The empty word is  $n = n + 0$ . An even letter substitutes  $x = J(x)^2 + \rho(x)$  into the inductive identity and lifts the new remainder through  $2^\ell$ . An odd letter cubes the identity and then substitutes  $x^3 = J(x)^2 + \rho(x)$ . Each step adds a nonnegative power-gap.  $\square$

For a one-letter illustration,  $n = 3$  and  $w = O$  give  $J(3) = 5$  and  $3^3 = 27 = 5^2 + 2$ , so  $\Delta_O(3) = 2$ .

**Theorem 2.5 (vanishing).** If  $w$  is realized at  $n$ , the following are equivalent:  $\Delta_w(n) = 0$ ; every local remainder along  $w$  vanishes; and  $(J^{|w|}(n))^{2^{|w|}} = n^{3^{\#O(w)}}$ . In that case  $w$  is monochrome: either  $w = E^k$  and  $n = a^{2^k}$  for an even  $a$ , or  $w = O^k$  and  $n = a^{2^k}$  for an odd  $a$ . A realized mixed word therefore has  $\Delta_w(n) > 0$ .

*Proof.* Theorem 2.4 gives the first and third items. A power-gap vanishes if and only if its addend vanishes, so zero defect forces every remainder to vanish, and conversely. Vanishing remainders preserve parity, hence monochrome itineraries, and unique factorization produces the two power towers.  $\square$

**Corollary 2.7 (cycle surplus).** If  $w$  is a cycle word at  $n$ , then  $\Delta_w(n) = n^{3^{\#O(w)}} - n^{2^{|w|}}$  exactly.

*Proof.* Theorem 2.4 with  $m = n$ .  $\square$

A mixed word has strict total defect, but the relative slack of a single letter tends to 0 with the state, so no uniform local tax exists.

### 3. Inverse cells and the census

The exact one-step fibers are

$$J(n) = q \iff q^2 \leq n < (q+1)^2 \quad (n \text{ even})$$

and

$$J(n) = m \iff m^2 \leq n^3 < (m+1)^2 \quad (n \text{ odd}).$$

**Lemma 3.1 (odd cells are unique).** An odd fiber contains at most one integer. An even fiber is a parity-restricted square interval and may contain many predecessors.

*Proof.* Suppose  $a < b$  lie in the same odd cell indexed by  $m$ . Then  $(a+1)^3 \leq b^3 < (m+1)^2$  and  $m^2 \leq a^3$ . Subtracting the latter lower bound from the former upper bound gives

$$3a^2 + 3a + 1 = (a+1)^3 - a^3 < (m+1)^2 - m^2 = 2m + 1,$$

hence  $3a^2 + 3a < 2m$  and  $3a^2 < 2m$ . If  $a = 0$ , then  $m = 0$  and the displayed strict inequality is already impossible. If  $a > 0$ , squaring and using  $m^2 \leq a^3$  gives  $9a^4 < 4m^2 \leq 4a^3$ , contradicting  $4a^3 < 9a^4$ .  $\square$

**Theorem 3.2 (cycle restrictions).** Let  $w$  be a cycle word at  $n \geq 2$ .

- (i) The word is formally expanding:

$$2^{|w|} < 3^{\#O(w)}.$$

A contracting word cannot close a nontrivial cycle.

- (ii) The cycle minimum is odd and the cycle maximum is even. A minimum-based orientation cannot end in an odd letter.

- (iii) A realized word  $v$  is *superquadratic* if

$$3^{\#O(v)} \geq 2^{|v|+1}.$$

Any realized path from a start  $n \geq 2$  to a state at least  $n^2$  is superquadratic. On a cycle minimum the path to any later even state is superquadratic. The prefix *OOE* is expanding ( $9 > 8$ ) but not superquadratic ( $9 < 16$ ), so it cannot carry the minimum to square scale.

*Proof.* For (i), Theorem 2.2 applied to a cycle endpoint gives  $n^{2^{|w|}} \leq n^{3^{\#O(w)}}$ . Since  $n \geq 2$ , comparison of exponents gives  $2^{|w|} \leq 3^{\#O(w)}$ ; equality is impossible because the two sides have different prime divisors and  $|w| \geq 1$ . Alternatively: a mixed cycle word has  $\Delta_w(n) > 0$  by Theorem 2.5, so the envelope is strict; a monochrome tower cannot return for  $n \geq 2$ .

Every state on such a cycle is at least 2: once an orbit reaches 1, it remains there and cannot return to a start  $n \geq 2$ . An even state  $x \geq 2$  satisfies  $J(x) < x$ , so a cycle minimum cannot be even. An odd state  $x \geq 3$  satisfies  $J(x) > x$ , so a cycle maximum cannot be odd. This proves the first assertion of (ii). For the final-letter assertion, let  $x$  be the predecessor of a minimum-oriented return  $n$ . If the last letter were odd, the odd return cell would give  $n^2 \leq x^3 < (n+1)^2$ . Minimality gives  $x \geq n$ , hence  $n^3 < (n+1)^2$ , impossible for  $n \geq 3$ ; and the minimum is odd, so  $n \geq 3$ .

For (iii), let a realized word  $v$  send  $n$  to  $y \geq n^2$ . Theorem 2.2 gives

$$n^{2^{|v|+1}} = (n^2)^{2^{|v|}} \leq y^{2^{|v|}} \leq n^{3^{\#O(v)}}.$$

Since  $n \geq 2$ , comparison of exponents proves the superquadratic inequality. On a cycle minimum, if a later state  $y$  is even, its successor is at least  $n$ . Thus  $n \leq J(y) = \lfloor \sqrt{y} \rfloor$ , so  $n^2 \leq y$ , and the preceding argument applies to that prefix.  $\square$

**Lemma 3.3 (coarse lower envelope).** For  $n \geq 1$ , write  $q = \lfloor \sqrt{n} \rfloor$ . Then  $q^2 \leq n < (q+1)^2$ . For  $q \geq 1$  one has  $(q+1)^2 \leq 4q^2$ , and the second inequality is strict for  $q \geq 2$ , while for  $q = 1$  one has  $n < 4$ . In all cases

$$n < 4 \lfloor \sqrt{n} \rfloor^2.$$

Thus an even step satisfies  $n \leq 4J(n)^2$  and an odd step satisfies  $n^3 \leq 4J(n)^2$ . Composing along a realized word  $v$  of length  $k$  with odd count  $o$  gives

$$n^{3^o} \leq C_v J^k(n)^{2^k}.$$

The constant starts at 1 and updates by  $C \mapsto C \cdot 4^{2^j}$  on an even letter and  $C \mapsto C^3 \cdot 4^{2^j}$  on an odd letter, at step  $j$ . In particular  $C_{OOO} = 2^{38}$  and  $C_{OOOO} = 2^{130}$ .

*Proof.* The comparison  $n < 4q^2$  is the paragraph above. The composition law is the same recurrence as the proof of Theorem 2.2, with a factor  $4^{2^j}$  inserted at each letter. The values  $C_{OOO}$  and  $C_{OOOO}$  are the result of that recurrence on those two words.  $\square$

**Lemma 3.4 (next-square thresholds).** (i) If  $q \geq 5$  realizes *OO*, then  $J^2(q) \geq (q+1)^2$ . (ii) If  $q \geq 3$  realizes *OOO*, then  $J^3(q) \geq (q+1)^2$ . (iii) If a realized

word  $v$  satisfies  $J^{|v|}(q) \geq (q+1)^2$  and the next realized letter is odd, then  $J^{|v|+1}(q) \geq (q+1)^2$ . (iv) If  $vE$  is a cycle word at  $n$ , then  $J^{|v|}(n) < (n+1)^2$ . (v) If  $a \geq 3$ , then  $O^a E$  is not a cycle word at any  $n \geq 2$ .

*Proof.* For (i), write  $m = \lfloor q^{3/2} \rfloor$ . The bound  $J^2(q) \geq (q+1)^2$  follows from  $m^3 \geq (q+1)^4$ , because then  $\lfloor m^{3/2} \rfloor \geq (q+1)^2$ . If  $q = 5$ , then  $11^2 = 121 \leq 125 = 5^3 < 144 = 12^2$ , so  $m = 11$ , and  $11^3 = 1331 \geq 6^4 = 1296$ . Since  $q$  realizes  $OO$ , it is odd, so the only remaining case is  $q \geq 7$ . Then  $m \geq q^{3/2} - 1$ , so it is enough that  $(q^{3/2} - 1)^3 \geq (q+1)^4$ . Expanding the left side and dropping the positive remainder  $3q^{3/2} - 1$  reduces this to  $q^{9/2} - 3q^3 \geq (q+1)^4$ , or equivalently  $\sqrt{q} - 3/q \geq (1 + 1/q)^4$ . The left side increases for  $q > 0$  and the right side decreases, so the case  $q = 7$  suffices:

$$\sqrt{7} - \frac{3}{7} > \frac{5}{2} - \frac{3}{7} = \frac{29}{14} > \frac{4096}{2401} = \left(\frac{8}{7}\right)^4,$$

where  $\sqrt{7} > 5/2$  follows from  $25/4 < 7$ .

For (ii), the orbit  $3 \rightarrow 5 \rightarrow 11 \rightarrow 36$  gives  $J^3(3) = 36 \geq 16$ . If  $q \geq 5$  realizes  $OOO$ , then it realizes  $OO$ , so (i) gives  $J^2(q) \geq (q+1)^2$ . The third letter is odd, hence  $J^3(q) = J(J^2(q)) \geq J^2(q)$ .

For (iii), the image after  $v$  is odd and at least  $(q+1)^2 \geq 4$ , so the odd branch does not decrease it.

For (iv), the last letter is even, so the preimage  $z = J^{|v|}(n)$  is even and satisfies  $n^2 \leq z < (n+1)^2$ .

For (v), suppose  $O^a E$  is a cycle word at  $n \geq 2$ . The start is odd, hence at least 3, and realizes  $O^a$ . Parts (ii) and (iii) give  $J^a(n) \geq (n+1)^2$ , contradicting (iv).  $\square$

**Lemma 3.5 (two length-six exclusions).** Neither  $OOOEEOE$  nor  $OOOOEE$  is a cycle word at any  $n \geq 2$ .

*Proof.* First, if  $n \geq 256$ , then

$$n^{81} > 2^{130}(n+1)^{64}.$$

Indeed  $257^{64} < 2 \cdot 256^{64}$  because  $(1 + 1/256)^{64} < e^{64/256} = e^{1/4} < 2$ , and  $256(n+1) \leq 257n$ , so  $(n+1)^{64} < 2n^{64}$ . Then  $2^{130}(n+1)^{64} < 2^{131}n^{64}$ . Since  $n \geq 256 = 2^8$ , one has  $n^{17} \geq 2^{136}$ , and therefore  $2^{131}n^{64} < 2^{136}n^{64} \leq n^{81}$ .

For  $2 \leq n < 256$ , neither word returns to its start. This is a table of 254 evaluations of each word: at every start, either some letter fails to match the current parity, or the six-step image differs from the start. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_oooeoe` and `no_cycle_word_ooooee` (Appendix A).

Now suppose  $OOOOEE$  is a cycle word at  $n \geq 256$ , and write  $z = J^4(n)$  for the image after the prefix  $OOOO$ . Lemma 3.4(iv) gives  $J(z) < (n+1)^2$ , and

the preceding even letter gives  $z < (J(z) + 1)^2$ , hence  $z < (n + 1)^4$ . Lemma 3.3 on  $OOOO$  gives  $n^{81} \leq 2^{130} z^{16}$ , so  $n^{81} < 2^{130} (n + 1)^{64}$ , contradicting the tail inequality.

Finally suppose  $OOOEOE$  is a cycle word at  $n \geq 256$ . Write  $z_3 = J^3(n)$  and  $y = J(z_3) = \lfloor \sqrt{z_3} \rfloor$ , so  $z_3 < (y + 1)^2$ . Lemma 3.3 on  $OOO$  gives  $n^{27} \leq 2^{38} z_3^8 < 2^{38} (y + 1)^{16}$ . Cubing yields  $n^{81} < 2^{114} (y + 1)^{48}$ . The last letters  $OE$  give the odd-cell bound  $y^3 < (n + 1)^4$ . Write  $A = n + 1 \geq 257$ . We claim  $(y + 1)^3 < 2A^4$ . If  $y \leq A$ , this is  $(A + 1)^3 < 2A^4$ . If  $y > A$ , then  $y \geq 258$ , so  $3y + 1 < y^2$  and hence  $(y + 1)^3 = y^3 + 3y^2 + 3y + 1 < A^4 + 4y^2$ , while  $4y^2 < A^4$  because  $4y^3 < 4A^4 \leq yA^4$ . Raising  $(y + 1)^3 < 2A^4$  to the sixteenth power gives  $(y + 1)^{48} < 2^{16} (n + 1)^{64}$ . Combining with the cubed lower envelope produces again  $n^{81} < 2^{130} (n + 1)^{64}$ .  $\square$

**Theorem 3.6 (small-cycle census).** No word of length at most six is a cycle word at any  $n \geq 2$ . Equivalently, a nontrivial Juggler cycle, if one exists, has period at least seven.

*Proof.* Rotating a cycle word by one letter moves the base point one step along the orbit and yields another cycle word; every state of the cycle is at least 2, because an orbit that reaches 1 stays at 1 and cannot return to a start  $n \geq 2$ .

If every letter is odd, the start is odd, hence  $n \geq 3$ , and the odd branch strictly increases there:  $J(x) > x$  for odd  $x \geq 3$ , since  $x^3 \geq (x + 1)^2$ . The orbit ascends strictly and never returns. Otherwise some letter is even, and a rotation ending just after that letter produces an even-terminating cycle word  $vE$  of the same length based at a cycle state  $m \geq 2$ . It therefore suffices to exclude even-terminating cycle words of length at most six.

By Theorem 3.2(i) a cycle word is formally expanding. No even-terminating word of length one or two is expanding ( $2 > 3^0$  and  $4 > 3$ ).

Length three: the only expanding candidate is  $OOE$ . If  $m \geq 5$  realizes  $OO$ , Lemma 3.4(i) gives  $J^2(m) \geq (m + 1)^2$ , contradicting Lemma 3.4(iv). The only smaller odd start is  $m = 3$ , where  $J^2(3) = 11$  is odd, so the final even letter is not realized.

Length four: the expanding filter requires three odd letters among the first three, leaving only  $O^3E$ , excluded by Lemma 3.4(v).

Length five: the filter requires four odd letters among the first four, leaving only  $O^4E$ , excluded by Lemma 3.4(v).

Length six: the filter requires at least four odd letters among the first five, leaving  $O^5E$ ,  $EOOOOE$ ,  $OEEOOE$ ,  $OOEOOE$ ,  $OOOEOE$ , and  $OOOOEE$ . Lemma 3.4(v) excludes  $O^5E$ . The word  $EOOOOE$  rotates one step onto  $OOOEOE$ , and  $OEEOOE$  rotates two steps onto  $OOOEOE$ ; both are excluded by Lemma 3.5.

It remains to exclude  $OOEOOE$ . Let this word be a cycle word at some start, and rotate to a cycle minimum  $m \geq 2$ . The three alignments of the two even

letters are  $OOEOOE$ ,  $OEOOEO$ , and  $EOOEOO$ . The last starts even, so it cannot be a minimum, by Theorem 3.2(ii). The middle starts  $OE$ : the first image is even and strictly below  $m^2$ , whereas every even state after a cycle minimum is at least  $m^2$ , as proved in Theorem 3.2(iii). Thus the minimum orientation is  $OOEOOE$ . In particular  $m$  is odd, so  $m \geq 3$ . The prefix  $OOE$  must be realized. If  $m = 3$ , then  $3 \rightarrow 5 \rightarrow 11$  and 11 is odd, so  $OOE$  is not realized. Hence  $m \geq 5$ . Write  $y = J^3(m)$  for the state after  $OOE$ . Then  $y \geq m$  by minimality, so  $y \geq 5$ . The suffix  $OO$  is realized at  $y$ , and Lemma 3.4(i) gives  $J^2(y) \geq (y+1)^2 \geq (m+1)^2$ . The last letter is even, so Lemma 3.4(iv) gives  $J^2(y) < (m+1)^2$ .  $\square$

Lemma 3.4(v) excludes every odd-run-then-even word  $O^a E$  with  $a \geq 3$ , of any length.

**Lemma 3.7 (two length-seven exclusions).** Neither  $OOOOEOE$  nor  $OOOOOEE$  is a cycle word at any  $n \geq 2$ .

*Proof.* First, if  $n \geq 14$ , then

$$n^{243} > 2^{422}(n+1)^{128}.$$

Indeed  $14(n+1) \leq 15n$ , so  $(n+1)^{128} \leq (15/14)^{128}n^{128}$ . The finite comparison  $2^{422}15^{128} < 14^{243}$  then yields  $2^{422}(n+1)^{128} < 14^{115}n^{128} \leq n^{243}$ .

For  $2 \leq n < 14$ , neither word is realized: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooeoe` and `no_cycle_word_oooooee` (Appendix A).

Now suppose  $OOOOOEE$  is a cycle word at  $n \geq 14$ , and write  $z = J^5(n)$  for the image after the prefix  $OOOOO$ . Lemma 3.4(iv) on the last even letter, together with the preceding even letter, gives  $z < (n+1)^4$ . Lemma 3.3 on  $OOOOO$  gives  $n^{243} \leq 2^{422}z^{32}$ , so  $n^{243} < 2^{422}(n+1)^{128}$ , contradicting the tail inequality.

Finally suppose  $OOOOEOE$  is a cycle word at  $n \geq 14$ . Write  $z_4 = J^4(n)$  and  $y = J(z_4) = \lfloor \sqrt{z_4} \rfloor$ , so  $z_4 < (y+1)^2$ . Lemma 3.3 on  $OOOO$  gives  $n^{81} \leq 2^{130}z_4^{16} < 2^{130}(y+1)^{32}$ . Cubing yields  $n^{243} < 2^{390}(y+1)^{96}$ . The last letters  $OE$  give the odd-cell bound  $y^3 < (n+1)^4$ . Write  $A = n+1 \geq 15$ . The same comparison  $(y+1)^3 < 2A^4$  as in Lemma 3.5 holds at this smaller scale. Raising to the thirty-second power gives  $(y+1)^{96} < 2^{32}(n+1)^{128}$ . Combining with the cubed lower envelope produces again  $n^{243} < 2^{422}(n+1)^{128}$ .  $\square$

**Theorem 3.8 (small-cycle census through length seven).** No word of length at most seven is a cycle word at any  $n \geq 2$ . Equivalently, a nontrivial Juggler cycle, if one exists, has period at least eight.

*Proof.* The reduction of Theorem 3.6 applies at every length: an all-odd word cannot return, and every mixed cycle word rotates to an even-terminating cycle word based at a cycle state  $m \geq 2$ . Lengths at most six are Theorem 3.6. It remains to exclude even-terminating cycle words of length seven.

By Theorem 3.2(i) a cycle word is formally expanding. The even-terminating expanding length-seven words are exactly  $O^6E$ ,  $EO^5E$ ,  $OEO^4E$ ,  $OOEO^3E$ ,  $O^3EO^2E$ ,  $O^4EOE$ , and  $O^5EE$ . Lemma 3.4(v) excludes  $O^6E$ . The word  $EO^5E$  rotates one step onto  $O^5EE$ , and  $OEO^4E$  starts  $OE$ , so it cannot be a cycle minimum (Theorem 3.2(iii)) and rotates two steps onto  $O^4EOE$ ; both leftovers are excluded by Lemma 3.7.

It remains to exclude  $OOEO^3E$  and  $O^3EO^2E$ . Rotate either word to a cycle minimum  $m \geq 2$ . For  $OOEO^3E$  the minimum orientation retains the internal even letter followed by the suffix  $OOO$ . Then  $m \geq 3$ , the prefix through that even letter is realized, and Lemma 3.4(ii) at threshold 3 contradicts the last even cell. For  $O^3EO^2E$  the same bootstrap applies with suffix  $OO$  and threshold 5, once  $m = 3$  is removed: at  $m = 3$  the state after  $OOO$  is even, so the next even letter is not realized.  $\square$

The census of Theorems 3.6 and 3.8 is a statement about period. The same cells organise leftover words by even-count. Write

$$e_a = 2(3^a - 2^a)$$

for  $a \geq 0$ .

**Lemma 3.9 (trailing even run).** If a cycle word based at  $n$  ends with  $r \geq 1$  even letters, the state immediately before that run is strictly less than  $(n+1)^{2^r}$ .

*Proof.* The case  $r = 1$  is Lemma 3.4(iv). Suppose the claim holds for some  $r \geq 1$ , and let  $vE^{r+1}$  be a cycle word at  $n$ . Write  $z = J^{|v|}(n)$ . The inductive hypothesis applied to the suffix  $E^r$  after the first of those even letters gives  $J(z) < (n+1)^{2^r}$ . The state  $z$  is even, so  $z < (J(z) + 1)^2 \leq ((n+1)^{2^r})^2 = (n+1)^{2^{r+1}}$ .  $\square$

**Lemma 3.10 (odd-run lower envelope).** If  $n \geq 1$  realizes  $O^a$ , then

$$n^{3^a} \leq 2^{e_a} J^a(n)^{2^a}.$$

*Proof.* Lemma 3.3 supplies a multiplicative constant  $C_v$  along any realized word, with  $C_\varepsilon = 1$ ,  $C \mapsto C \cdot 4^{2^j}$  on an even letter, and  $C \mapsto C^3 \cdot 4^{2^j}$  on an odd letter, at step  $j$ . On the pure odd word  $O^a$  every letter is odd, so  $C_{O^{a+1}} = C_{O^a}^3 \cdot 4^{2^a}$ . Writing  $C_{O^a} = 2^{e_a}$  with  $e_0 = 0$  yields the recurrence  $e_{a+1} = 3e_a + 2^{a+1}$ , because  $4^{2^a} = 2^{2^{a+1}}$ . The closed form  $e_a = 2(3^a - 2^a)$  satisfies the recurrence and the initial value.  $\square$

**Lemma 3.11 (seven-odd window).** No integer  $n$  with  $2 \leq n < 256$  realizes the word  $O^7$ .

*Proof.* This is a table of 254 seven-step evaluations: at every such start, some letter fails to match the current parity. The same finite check is the Lean `native_decide` evaluation behind `no_follows_seven_odds_of_lt256` (Appendix A).  $\square$

**Theorem 3.12 (two-even leftover families).** Let  $k \geq 6$  and  $n \geq 2$ . Neither  $O^{k-2}EE$  nor  $O^{k-3}EOE$  is a cycle word at  $n$ .

*Proof.* First, if  $n \geq 256$ , then

$$n^{3^{k-2}} > 2^{e_{k-2}}(n+1)^{2^k}.$$

The case  $k = 6$  is the tail inequality of Lemma 3.5. If the display holds at some  $k \geq 6$ , cubing both sides produces

$$n^{3^{k-1}} > 2^{3e_{k-2}}(n+1)^{3 \cdot 2^k}.$$

The recurrence of Lemma 3.10 gives  $e_{k-1} = 3e_{k-2} + 2^{k-1}$ , so the desired comparison at length  $k+1$  reduces to  $2^{2^{k-1}} < (n+1)^{2^k}$ . Equivalently  $2 < (n+1)^2$ , which holds for every  $n \geq 256$ .

Now suppose  $O^{k-2}EE$  is a cycle word at such an  $n$ . Write  $z = J^{k-2}(n)$ . Lemma 3.9 with  $r = 2$  gives  $z < (n+1)^4$ . Lemma 3.10 on the prefix  $O^{k-2}$  gives  $n^{3^{k-2}} \leq 2^{e_{k-2}}z^{2^{k-2}}$ , hence  $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$ , contradicting the tail.

Finally suppose  $O^{k-3}EOE$  is a cycle word at such an  $n$ . Write  $z = J^{k-3}(n)$  and  $y = \lfloor \sqrt{z} \rfloor$ , so  $z < (y+1)^2$ . Lemma 3.10 on  $O^{k-3}$  and cubing produce  $n^{3^{k-2}} < 2^{3e_{k-3}}(y+1)^{3 \cdot 2^{k-2}}$ . The last letters  $OE$  give the odd-cell bound  $y^3 < (n+1)^4$ . The comparison  $(y+1)^3 < 2(n+1)^4$  of Lemma 3.5 applies at this scale. Raising it to the power  $2^{k-2}$  and using  $e_{k-2} = 3e_{k-3} + 2^{k-2}$  recovers again  $n^{3^{k-2}} < 2^{e_{k-2}}(n+1)^{2^k}$ .

For  $2 \leq n < 256$ , the cases  $k = 6$  and  $k = 7$  are Lemmas 3.5 and 3.7. The remaining short words  $O^6EE$ ,  $O^5EOE$ , and  $O^6EOE$  fail to return on the same 254-start window; this is the Lean `native_decide` evaluation behind `no_cycle_word_two_even_ee` and `no_cycle_word_two_even_eoe` (Appendix A). Any longer leftover of either family begins with seven consecutive odd letters, which Lemma 3.11 forbids on this window.  $\square$

A cycle word  $w$  at  $n$  is *minimum-based* when  $n$  is a cycle minimum:  $J^j(n) \geq n$  for every  $0 \leq j < |w|$ . The remainder after a proper prefix of a cycle word need not itself be a cycle word at the prefix endpoint. The next statement therefore transports the tail inequality of Theorem 3.12, not the cycle-word exclusion at a later start.

**Theorem 3.13 (first-even transport).** Let  $n \geq 2$ . No minimum-based cycle word at  $n$  has the form  $O^aEO^bEE$  with  $a \geq 2$  and  $b \geq 4$ , or the form  $O^aEO^bEOE$  with  $a \geq 2$  and  $b \geq 3$ .

*Proof.* Write  $y = J^{a+1}(n)$  for the state after the first even letter. Minimum-basedness gives  $y \geq n$ . In the first family the remainder after that letter is  $O^bEE$  with  $b+2 \geq 6$ ; in the second it is  $O^bEOE$  with  $b+3 \geq 6$ . The trailing-even and last-odd cells of those remainders are measured against the cycle start  $n$ . Combined with Lemma 3.10 at the remainder start  $y$ , the same algebra as

in Theorem 3.12 produces

$$y^{3^{\ell-2}} < 2^{e_{\ell-2}}(n+1)^{2^\ell} \leq 2^{e_{\ell-2}}(y+1)^{2^\ell},$$

where  $\ell$  is the remainder length. If  $y \geq 256$ , the first paragraph of Theorem 3.12 supplies the opposite inequality at  $y$ .

If  $y < 256$ , then  $n \leq y < 256$ . A gapped leftover of total length at least 17 has  $a \geq 7$  or  $b \geq 7$ , so either the prefix or the remainder realizes seven consecutive odd letters, contradicting Lemma 3.11. The finitely many short-gap words with  $2 \leq a \leq 6$  and  $b \leq 6$  fail to be minimum-based cycle words on the window  $2 \leq n < 256$ ; this is the Lean `native_decide` evaluation behind `no_cycleMin_gapped_three_even_ee` and `no_cycleMin_gapped_three_even_eoe` (Appendix A).  $\square$

The hypothesis that the start is a cycle minimum is essential. If  $y < n$ , the leftover cell is measured against a larger start and need not contradict the tail at  $y$ . In particular, Theorem 3.13 does not assert that those words fail to be cycle words at a non-minimum start. That upgrade is Theorem 3.21.

**Theorem 3.14 (three trailing evens).** Let  $a \geq 6$  and  $n \geq 2$ . The word  $O^a EEE$  is not a cycle word at  $n$ .

*Proof.* Write  $z = J^a(n)$ . Lemma 3.9 with  $r = 3$  gives  $z < (n+1)^8$ . Lemma 3.10 then yields  $n^{3^a} < 2^{e_a}(n+1)^{2^{a+3}}$  on any such cycle word. For  $n \geq 128$  the opposite comparison

$$n^{3^a} > 2^{e_a}(n+1)^{2^{a+3}}$$

holds. The case  $a = 6$  is  $n^{729} > 2^{1330}(n+1)^{512}$ . Indeed, for  $n \geq 128$  one has  $(n+1)^{512} < (129/128)^{512}n^{512} < e^4 n^{512} < 64n^{512}$ , so the claimed bound reduces to  $n^{217} > 2^{1336}$ . Since  $n \geq 128 = 2^7$ , the left side is at least  $2^{1519}$ . Cubing the  $a = 6$  comparison produces the general case: the recurrence of Lemma 3.10 reduces the inductive step to  $(n+1)^4 > 2$ , which holds for every  $n \geq 128$ .

For  $2 \leq n < 128$ , the case  $a = 6$  is a table of evaluations of `OOOOOOEEE`: at every such start the word fails to return. The same finite check is the Lean `native_decide` evaluation behind `no_cycle_word_ooooooooee` (Appendix A). For  $a \geq 7$  the prefix contains seven consecutive odd letters, which Lemma 3.11 forbids on this window.  $\square$

**Theorem 3.15 (mixed bunched family EOEE).** Let  $a \geq 5$  and  $n \geq 2$ . The word  $O^a EOEE$  is not a cycle word at  $n$ .

*Proof.* First let  $n \geq 4$ , and write  $z = J^a(n)$ ,  $y = \lfloor \sqrt{z} \rfloor$ , and  $p = J(y)$ . Lemma 3.9 with  $r = 2$  after the prefix  $O^a EO$  gives  $p < (n+1)^4$ . The letter after  $y$  is odd, so Lemma 3.10 at length one yields  $y^3 \leq 4p^2 < 4(n+1)^8$ . For  $n \geq 4$  one has  $4(n+1)^8 < (n+1)^9$ , hence  $y < (n+1)^3$ . The even cell at  $z$  then gives  $z < (y+1)^2 \leq (n+1)^6$ . Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

For  $a = 5$  and  $n \geq 314$ , the opposite comparison  $n^{243} > 2^{422}(n+1)^{192}$  holds. The base instance is the finite inequality  $2^{422}315^{192} < 314^{243}$ . If the display holds at some  $n \geq 1$ , the elementary comparison  $n(n+2) < (n+1)^2$  upgrades it to the same display at  $n+1$ . Cubing then produces the comparison at  $a+1$ , once  $(n+1)^6 > 4$ . In particular the case  $a = 6$  already holds for every  $n \geq 16$ .

For  $2 \leq n < 314$  and  $a = 5$ , and for  $2 \leq n < 16$  and  $a = 6$ , the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eooo` (Appendix A). For  $a \geq 7$  and  $n < 256$ , Lemma 3.11 applies. For  $a \geq 6$  and  $n \geq 16$ , the tail of the previous paragraph applies.  $\square$

**Theorem 3.16 (mixed bunched family *EOOEE*).** Let  $a \geq 4$  and  $n \geq 2$ . The word  $O^a EOOEE$  is not a cycle word at  $n$ .

*Proof.* First let  $n \geq 32$ , and write  $z = J^a(n)$ ,  $y = \lfloor \sqrt{z} \rfloor$ , and  $p$  for the image after the prefix  $O^a EOO$ . Lemma 3.9 with  $r = 2$  gives  $p < (n+1)^4$ . The two letters after  $y$  are odd, so Lemma 3.10 at length two yields  $y^9 \leq 2^{10}p^4 < 2^{10}(n+1)^{16}$ . For  $n \geq 32$  one has  $2^{10} < (n+1)^2$ , hence  $y < (n+1)^2$ . The even cell at  $z$  then gives  $z < (y+1)^2 \leq (n+1)^4$ . Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For  $n \geq 256$  and  $a \geq 4$ , this is the opposite of the shared tail of Theorem 3.12 at length  $k = a + 2$ .

For  $2 \leq n < 256$ , the cases  $a = 4, 5, 6$  fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoooo` (Appendix A). For  $a \geq 7$ , Lemma 3.11 applies.  $\square$

**Theorem 3.17 (mixed bunched family *EOOOEE*).** Let  $a \geq 3$  and  $n \geq 2$ . The word  $O^a EOOOEE$  is not a cycle word at  $n$ .

*Proof.* First let  $a \geq 4$  and  $n \geq 3$ , and write  $z = J^a(n)$ ,  $y = \lfloor \sqrt{z} \rfloor$ , and  $p$  for the image after the prefix  $O^a EOOO$ . Lemma 3.9 with  $r = 2$  gives  $p < (n+1)^4$ . The three letters after  $y$  are odd, so Lemma 3.10 at length three yields  $y^{27} \leq 2^{38}p^8 < 2^{38}(n+1)^{32}$ . For  $n \geq 3$  one has  $2^{38} < (n+1)^{22}$ , hence  $y < (n+1)^2$ . The even cell at  $z$  then gives  $z < (y+1)^2 \leq (n+1)^4$ . Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For  $n \geq 256$  and  $a \geq 4$ , this is the opposite of the shared tail of Theorem 3.12 at length  $k = a + 2$ , already used in Theorem 3.16.

Now let  $a = 3$  and  $n \geq 256$ . The same two-even cell gives  $z < (y+1)^2$ , so Lemma 3.10 at the prefix  $O^3$  yields  $n^{27} < 2^{38}(y+1)^{16}$ . The three-odd envelope on  $y$  still gives  $y^{27} < 2^{38}(n+1)^{32}$ .

If  $y < 39$ , then  $n^{27} < 2^{38} \cdot 39^{16}$ . The numerical comparison  $2^{38} \cdot 39^{16} < 24^{27}$  contradicts  $n \geq 24$ .

If  $y \geq 39$ , the numerical comparison  $40^{27} < 2 \cdot 39^{27}$  upgrades to  $(y+1)^{27} < 2y^{27}$ , hence  $(y+1)^{27} < 2^{39}(n+1)^{32}$ . Cubing the prefix bound three times produces  $n^{729} < 2^{1026}(y+1)^{432}$ . Raising the successor bound to the sixteenth power produces  $(y+1)^{432} < 2^{624}(n+1)^{512}$ . Combining these displays yields  $n^{729} < 2^{1650}(n+1)^{512}$ . The opposite comparison holds at  $n = 197$  and persists to every larger start by the elementary comparison  $n(n+2) < (n+1)^2$  already used in Theorem 3.15.

For  $2 \leq n < 256$ , the cases  $a = 3, 4, 5, 6$  fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooooe` (Appendix A). For  $a \geq 7$ , Lemma 3.11 applies.  $\square$

**Theorem 3.18 (mixed bunched family  $EEOE$ ).** Let  $a \geq 5$  and  $n \geq 2$ . The word  $O^a EEOE$  is not a cycle word at  $n$ .

*Proof.* First let  $n \geq 4$ , and write  $z = J^a(n)$  and  $y$  for the last odd letter of  $O^a EEOE$ . The suffix  $EOE$  is a cycle suffix, so the last-odd cube of Theorem 3.15 gives  $y^3 < (n+1)^4$ . The two letters between  $z$  and  $y$  are even, so  $z < (y+1)^4$ . For  $n \geq 4$  the successor comparison  $(y+1)^3 < 2(n+1)^4$  upgrades this to  $z < (n+1)^6$ . Combined with Lemma 3.10, any such cycle word would satisfy the same display as Theorem 3.15:

$$n^{3^a} < 2^{e_a}(n+1)^{6 \cdot 2^a}.$$

The opposite comparison is therefore the tail of Theorem 3.15: it holds for  $a = 5$  and  $n \geq 314$ , and already for  $a = 6$  and  $n \geq 16$ .

For  $2 \leq n < 314$  and  $a = 5$ , and for  $2 \leq n < 16$  and  $a = 6$ , the word fails to return; these are the Lean `native_decide` evaluations behind `no_cycle_word_three_even_eeoe` (Appendix A). For  $a \geq 7$  and  $n < 256$ , Lemma 3.11 applies. For  $a \geq 6$  and  $n \geq 16$ , the tail of the previous paragraph applies.  $\square$

**Theorem 3.19 (mixed bunched family  $EOEOE$ ).** Let  $a \geq 4$  and  $n \geq 2$ . The word  $O^a EOEOE$  is not a cycle word at  $n$ .

*Proof.* First let  $n \geq 32$ , and write  $z = J^a(n)$ ,  $w = \lfloor \sqrt{z} \rfloor$ , and  $y$  for the last odd letter. The suffix  $EOE$  again gives  $y^3 < (n+1)^4$ . The one-odd envelope on  $w$  yields  $w^3 \leq 4s^2$ , where  $s$  is the image after  $O^a EO$ . The last-odd cell and  $n \geq 32$  upgrade this to  $w < (n+1)^2$ , hence  $z < (w+1)^2 \leq (n+1)^4$ . Combined with Lemma 3.10, any such cycle word would satisfy

$$n^{3^a} < 2^{e_a}(n+1)^{2^{a+2}}.$$

For  $n \geq 256$  and  $a \geq 4$ , this is the shared tail already used in Theorem 3.16.

For  $2 \leq n < 256$ , the cases  $a = 4, 5, 6$  fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eoeoe` (Appendix A). For  $a \geq 7$ , Lemma 3.11 applies.  $\square$

**Theorem 3.20 (mixed bunched family  $EOOEOE$ ).** Let  $a \geq 3$  and  $n \geq 2$ . The word  $O^aEOOEOE$  is not a cycle word at  $n$ .

*Proof.* First let  $a \geq 4$  and  $n \geq 4$ , and write  $z = J^a(n)$ ,  $u = \lfloor \sqrt{z} \rfloor$ , and  $y$  for the last odd letter. The suffix  $EOE$  gives  $y^3 < (n+1)^4$ , hence  $(y+1)^3 < 2(n+1)^4$ . The two letters after  $u$  are odd, so Lemma 3.10 at length two yields  $u^9 \leq 2^{10}s^4 < 2^{10}(y+1)^8$ . Cubing that display against the last-odd successor bound produces  $u^{27} < 2^{38}(n+1)^{32}$ . The same comparison as in Theorem 3.17 then gives  $u < (n+1)^2$ , hence  $z < (u+1)^2 \leq (n+1)^4$ . Combined with Lemma 3.10, any such cycle word would satisfy the shared two-even tail of Theorem 3.16.

Now let  $a = 3$  and  $n \geq 256$ . The prefix  $O^3$  against  $z < (u+1)^2$  yields  $n^{27} < 2^{38}(u+1)^{16}$ , and the two-odd plus last-odd geometry of the previous paragraph yields  $u^{27} < 2^{38}(n+1)^{32}$ . These are the same two displays as in the  $a = 3$  case of Theorem 3.17, with  $u$  in place of  $y$ , and the same small/large split applies.

For  $2 \leq n < 256$ , the cases  $a = 3, 4, 5, 6$  fail to return on that window; this is the Lean `native_decide` evaluation behind `no_cycle_word_three_even_eooeee` (Appendix A). For  $a \geq 7$ , Lemma 3.11 applies.  $\square$

**Theorem 3.21 (gapped leftovers as cycle words).** Let  $n \geq 2$ . No cycle word at  $n$  has the form  $O^aEO^bEE$  with  $a \geq 2$  and  $b \geq 4$ , or the form  $O^aEO^bEOE$  with  $a \geq 2$  and  $b \geq 3$ .

*Proof.* Every cycle word has a minimum-based rotation. It is therefore enough to check that every cyclic shift of either word is an already-excluded cycle-minimum orientation.

Write  $w$  for the gapped word. In the first family,  $|w| = a + b + 3$ . The rotation by  $k = 0$  is the original word, excluded as a cycle minimum by Theorem 3.13. The rotation by  $k = a + 1$  is the bootstrap word  $O^bEEO^aE$ . That word has an internal even letter and last gap at least 2. If  $a \geq 3$ , the last-gap threshold of Lemma 3.4 at  $OOO$  and  $N = 3$  excludes it. If  $a = 2$ , the same lemma at  $OO$  and  $N = 5$  excludes every start  $n \geq 5$ ; the remaining odd start  $n = 3$  does not realize four consecutive odd letters, so it cannot follow  $O^b$  for  $b \geq 4$ . The rotation by  $k = a + b + 2$  begins with the last even letter of  $w$ , which Theorem 3.2 forbids at a cycle minimum. Every other rotation ends with an odd letter, likewise forbidden at a cycle minimum.

In the second family,  $|w| = a + b + 4$ . The same four classes appear: the original word is Theorem 3.13; the rotation by  $k = a + 1$  is  $O^bEOEO^aE$ , excluded by the same last-gap thresholds, with the remaining start  $n = 3$  and  $a = 2$  either failing to realize four odds or, when  $b = 3$ , reaching 6 after  $OOOE$  and then meeting an odd letter; the rotation by  $k = a + b + 2$  begins  $OE$ , forbidden by Theorem 3.2; and every other rotation ends odd.

The original start need not be a cycle minimum. After rotation the start is a minimum, so the hypothesis  $y < n$  that blocked Theorem 3.13 does not arise.  $\square$

Theorems 3.12–3.21 assemble into an even-count exclusion: no cycle word has fewer than four even letters, so a nontrivial cycle has period at least eleven (Theorem 3.22). Section 4 excludes later periods by financing.

A cycle minimum starts with two odd letters, ends with an even letter, and is formally expanding (Theorem 3.2). Every integer  $2 \leq n < 12$  reaches 1, so a cycle minimum is at least 12. Write  $e$  for the number of even letters.

**Lemma 3.21b (canonical run form).** After rotation to a minimum-based orientation, the word begins with  $OO$  and ends with  $E$ ; hence every word with  $1 \leq e \leq 3$  even letters has the canonical run decomposition  $O^aEO^bEO^cE$  (unused runs empty). The case  $e = 0$  is all-odd and is already forbidden by the last-letter restriction. No other cyclic rotation needs a separate case.

*Proof.* Theorem 3.2: the minimum is odd, so the word cannot begin with  $E$  or  $OE$ , and it cannot end with an odd letter. Thus a minimum-based word starts  $OO$  and ends  $E$ , so  $e \geq 1$ . The remaining letters are odd runs separated by the  $e$  even letters, so the word is  $O^{a_1}E \cdots O^{a_e}E$  with  $a_1 \geq 2$ . For  $e \leq 3$  this is  $O^aEO^bEO^cE$  with unused runs empty. Every cycle word has a minimum-based rotation of the same even-count, and that orientation is already in this form.  $\square$

**Lemma 3.21a (classification).** Every minimum-based cycle word with at most three even letters belongs to one of the families excluded by Lemma 3.4(v) and Theorems 3.12–3.21.

*Proof.* Lemma 3.21b puts the word in the form  $O^aEO^bEO^cE$  with  $a \geq 2$  and unused runs empty. If  $e = 0$ , the word is all-odd. If  $e = 1$ , it is  $O^aE$ . If  $e = 2$ , a last run  $c \geq 2$  is the internal-even bootstrap of Lemma 3.4, and the remaining shapes are the two-even families of Theorem 3.12. If  $e = 3$ , again  $c \geq 2$  is the bootstrap, while  $c \in \{0, 1\}$  is either a gapped leftover (Theorem 3.21) or one of the seven bunched families (Theorems 3.14–3.20).  $\square$

| $e$      | remaining<br>minimum-based                | elimination                    |
|----------|-------------------------------------------|--------------------------------|
|          | forms                                     |                                |
| 0        | all-odd                                   | cannot return<br>(Theorem 3.6) |
| 1        | $O^aE$                                    | next-square; Lemma<br>3.4(v)   |
| 2        | $O^aEE$ , $O^aEOE$ , last<br>run $\geq 2$ | Theorem 3.12;<br>Lemma 3.4     |
| 3        | bunched families and<br>gapped leftovers  | Theorems 3.14–3.21             |
| $\geq 4$ | not excluded by<br>even-count             | finance (Section 4)            |

**Theorem 3.22 (even-count).** No word with fewer than four even letters is a

cycle word at any  $n \geq 2$ . Equivalently, a nontrivial cycle word has at least four even letters.

*Proof.* Every cycle word has a minimum-based rotation, with the same even-count. Lemma 3.21b puts that orientation in canonical run form; Lemma 3.21a names the families. In detail:

If  $e = 0$ , the word is all-odd and cannot return, as in Theorem 3.6.

If  $e = 1$ , the word is  $O^a E$  with  $a \geq 2$ . The case  $a = 2$  is  $OOE$ , excluded in Theorem 3.6. The cases  $a \geq 3$  are Lemma 3.4(v).

If  $e = 2$ , the word is  $O^a EO^c E$  with  $a \geq 2$ . A last odd-run  $c \geq 2$  is an internal even letter followed by  $OO$  or  $OOO$ . Lemma 3.4 at the cycle minimum ( $n \geq 12$ ) contradicts the last-even cell. The remaining shapes are  $O^a EE$  and  $O^a EOE$ . Expansion forces  $a \geq 4$  and  $a \geq 3$  respectively, so both are Theorem 3.12.

If  $e = 3$ , the word is  $O^a EO^b EO^c E$  with  $a \geq 2$ . Again  $c \geq 2$  is the internal-even bootstrap. The remaining last runs are  $c = 0$  and  $c = 1$ . For  $c = 0$  and  $b \geq 4$  the word is a gapped leftover  $O^a EO^b EE$ , excluded by Theorem 3.21. For  $c = 0$  and  $b \leq 3$  the word is one of  $O^a EEE$ ,  $O^a EOOE$ ,  $O^a EOOEE$ ,  $O^a EOOOEE$ , excluded by Theorems 3.14–3.17 once expansion supplies the stated lower bounds on  $a$ . For  $c = 1$  and  $b \geq 3$  the word is a gapped leftover  $O^a EO^b EOE$ , excluded by Theorem 3.21. For  $c = 1$  and  $b \leq 2$  the word is one of  $O^a EEOE$ ,  $O^a EEOEOE$ ,  $O^a EOOEOE$ , excluded by Theorems 3.18–3.20.

Thus  $e \leq 3$  is impossible.  $\square$

**Corollary 3.23.** A nontrivial cycle, if one exists, has period at least eleven.

*Proof.* Theorem 3.22 gives four even letters, hence  $o \leq L - 4$ . Formal expansion is  $2^L < 3^o \leq 3^{L-4}$ . The comparison  $2^L < 3^{L-4}$  first holds at  $L = 11$ .  $\square$

In particular there is no cycle of length eight, nine, or ten. A single coarse successor power  $(n + 1)^K$  cannot exclude the four families of Theorems 3.17–3.20 by the same cell used for Theorems 3.15 and 3.16: at the first expanding prefix length, the exponent  $K \cdot 2^a$  meets or exceeds  $3^a$  for those four words, so those arguments use a tight last-odd cell.

## 4. Cycle finance

A cycle word is formally expanding (Theorem 3.2), yet the orbit returns exactly. The multiplicative surplus  $3^o - 2^L$  must be financed by the floor remainders, which are relatively  $O(1/x)$  in logarithms. The resulting bound on the cycle minimum excludes every period that is not admissible for the inequality, once a uniform logarithmic floor-error bound is available above a verified descent floor.

Throughout this section write  $L = |w|$  and  $o = \#O(w)$  for a cycle word  $w$  based at a cycle minimum  $n \geq 2$ . Natural logarithms are written  $\log$ . The unique

one-step fibres of Section 3 give, for every state  $x \geq 1$  with image  $y = J(x)$ ,

$$y^2 \leq x^e < (y+1)^2, \quad e = \begin{cases} 1, & x \text{ even}, \\ 3, & x \text{ odd}. \end{cases}$$

**Lemma 4.1 (dyadic-cell logarithm).** If  $z, y \geq 1$  and  $z < (y+1)^2$ , then  $\log z \leq 2 \log y + 2/y$ .

*Proof.* The hypothesis gives  $\log z \leq 2 \log(y+1)$ . The inequality  $\log(1+u) \leq u$  for  $u > 0$  yields  $\log(y+1) = \log y + \log(1+1/y) \leq \log y + 1/y$ .  $\square$

**Lemma 4.2 (one-step bounds).** Let  $x \geq 2$  and  $y = J(x)$ . If  $x$  is even, then  $\log x \leq 2 \log y + 2/y$ . If  $x$  is odd, then  $3 \log x \leq 2 \log y + 2/y$ .

*Proof.* The even cell is  $y^2 \leq x < (y+1)^2$ . The cell logarithm lemma on  $z = x$  is the first claim. The odd cell is  $y^2 \leq x^3 < (y+1)^2$ . The same lemma on  $z = x^3$  gives  $\log(x^3) \leq 2 \log y + 2/y$ .  $\square$

On a cycle minimum every proper prefix is non-contracting: a prefix with  $3^{\#O} < 2^k$  would satisfy  $J^k(n) < n$  by Corollary 2.3, contradicting minimality. Thus  $2^k \leq 3^{o_k}$  for every prefix of length  $k$ , where  $o_k$  is the odd count of that prefix.

**Lemma 4.3 (unrolled envelope).** Write  $x_k = J^k(n)$  and  $o_k$  for the odd count of the length- $k$  prefix. For every  $0 \leq k \leq L$ ,

$$3^{o_k} \log n \leq 2^k \log x_k + \frac{k 3^{o_k}}{n}.$$

*Proof.* The case  $k = 0$  is an equality. Suppose the claim holds at  $k < L$ , and write  $x = x_k$  and  $y = x_{k+1}$ . Minimality gives  $n \leq y$ , and the prefix law gives  $2^{k+1} \leq 3^{o_{k+1}}$ , hence

$$\frac{2^{k+1}}{y} \leq \frac{3^{o_{k+1}}}{n}.$$

If the next letter is even, then  $o_{k+1} = o_k$  and the one-step bound gives  $\log x \leq 2 \log y + 2/y$ . Multiply by  $2^k$  and add the inductive remainder:

$$3^{o_k} \log n \leq 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k 3^{o_k}}{n} \leq 2^{k+1} \log y + \frac{(k+1) 3^{o_k}}{n}.$$

If the next letter is odd, then  $o_{k+1} = o_k + 1$ . Multiply the inductive bound by 3 and apply the one-step bound in the form  $3 \log x \leq 2 \log y + 2/y$ :

$$3^{o_{k+1}} \log n \leq 2^{k+1} \log y + \frac{2^{k+1}}{y} + \frac{k 3^{o_k+1}}{n} \leq 2^{k+1} \log y + \frac{(k+1) 3^{o_{k+1}}}{n}.$$

This is the claim at  $k+1$ .  $\square$

**Theorem 4.4 (finance).** Let  $w$  be a cycle word of length  $L$  with  $o$  odd letters, based at a cycle minimum  $n \geq 2$ . Then

$$n \log n \cdot (3^o - 2^L) \leq L \cdot 3^o.$$

The rest of the note exists to establish this inequality and apply it at the verified descent floor.

*Proof.* Apply Lemma 4.3 at  $k = L$ . Periodicity gives  $x_L = n$  and  $o_L = o$ , so

$$3^o \log n \leq 2^L \log n + \frac{L \cdot 3^o}{n}.$$

The word is formally expanding, so  $3^o > 2^L$ . Rearranging and multiplying by  $n$  is the claim.  $\square$

This is the conceptual centre of the note. The left-hand side is the formal exponential surplus  $3^o - 2^L$  scaled by  $n \log n$ ; the right-hand side is the floor-error budget available from  $L$  steps. A cycle must expand under the idealized map, but the exact orbit returns, so the surplus must be paid by floor errors. At large state size those errors are too small to finance a non-negligible surplus. The only analytic input is  $\log(1 + u) \leq u$ . The Lean form is exactly Theorem 4.4 (constant 1).

This is a floor-power realization of a financing method, not a new method. The Juggler-specific form of the inequality, and the explicit period bounds it produces, appear to be new.

The computational table of Theorem 4.6 uses a weaker per-step bound, valid on every cycle because every start below 12 reaches 1 (so every cycle state is at least 12). For a state  $x$  with image  $y = J(x) \geq 12$ , the relative defect  $\delta = (x^e - y^2)/x^e$  satisfies  $\delta \leq 2/y \leq 1/6$ . Writing  $\varepsilon = -\frac{1}{2} \log(1 - \delta)$  and using  $-\log(1 - \delta) \leq \frac{6}{5} \delta$  on  $[0, 1/6]$  gives  $\varepsilon \leq (6/5)/y$ . Unrolling  $t_{i+1} = (e_i/2)t_i - \varepsilon_i$  around the cycle then yields

$$1 - \frac{2^L}{3^o} \leq \frac{6}{5} \sum_{i=1}^L \frac{1}{x_i \log x_i} \leq \frac{6}{5} \cdot \frac{L}{n \log n}.$$

This is Theorem 4.4 with an extra factor  $6/5$  on the right, so it is conservative relative to constant 1. We use the weaker  $6/5$  bound because it is uniform above the verified descent floor and suffices to obtain the stated 1054 threshold; no optimization of the constant is attempted.

The right-hand side is largest at the least admissible odd count  $o_{\min}(L) = \min\{o : 3^o > 2^L\}$ . Define

$$\gamma(L) = \frac{3^{o_{\min}(L)}}{2^L} - 1, \quad B(L) = \frac{6}{5} \cdot \frac{L \cdot 3^{o_{\min}(L)}}{3^{o_{\min}(L)} - 2^L}, \quad n_{\max}(L) = \max\{n \in \mathbb{N} : n \log n \leq B(L)\}.$$

The quantity  $\gamma(L)$  is the one-sided relative gap of  $L \log 2$  to the next multiple of  $\log 3$ . It is small precisely when  $o_{\min}/L$  is a good one-sided approximation

to  $\log 2/\log 3$ . The record minima of  $\gamma(L)$  form a sequence of one-sided best-approximation lengths related to the continued-fraction structure of  $\log 2/\log 3$ ; they include

$$1, 3, 11, 19, 84, 569, 1054, \dots$$

They are not, as a list, the ordinary continued-fraction convergent denominators of that ratio. They are also not the whole admissible set: multiples and sums of those lengths also make  $\gamma$  small.

**Proposition 4.4a (exceptional-length algorithm).** Fix a verified descent floor  $N_0$ . For each integer  $1 \leq L \leq 10^5$ , compute  $o_{\min}(L) = \min\{o : 3^o > 2^L\}$  and  $n_{\max}(L) = \max\{n \in \mathbb{N} : n \log n \leq B(L)\}$  by exact integer arithmetic, and retain  $L$  if and only if  $n_{\max}(L) > N_0$ . The resulting set is

$$\mathcal{E}(N_0) = \{L : 1 \leq L \leq 10^5, n_{\max}(L) > N_0\}.$$

A length lies in  $\mathcal{E}(N_0)$  if and only if it is *admissible to this bound*: Corollary 4.5 does not exclude it at that floor. The set is not the set of dynamically plausible periods, and membership is not evidence for a cycle. The printed instance is  $\mathcal{E} = \mathcal{E}(10^6)$ , with  $|\mathcal{E}| = 397$ .

**Corollary 4.5.** If every integer  $2 \leq n \leq N_0$  reaches 1, then no nontrivial cycle of length  $L$  exists whenever  $n_{\max}(L) \leq N_0$ .

*Proof.* A periodic state never reaches 1, so every cycle state is at least  $N_0 + 1$ . Theorem 4.4 in the  $6/5$  form forces the minimum to satisfy  $n \log n \leq B(L)$ , hence  $n \leq n_{\max}(L)$ .  $\square$

Record values include  $n_{\max}(19) = 297$ ,  $n_{\max}(84) = 5599$ ,  $n_{\max}(569) = 58398$ , and  $n_{\max}(1054) = 1997197$ .

**Theorem 4.6 (verified computation).** Every integer  $2 \leq n \leq 10^6$  reaches 1. Consequently:

- (A) there is no nontrivial Juggler cycle of length at most 1053;
- (B) if a nontrivial cycle has period  $L \leq 10^5$ , then  $L \in \mathcal{E}$ , where  $|\mathcal{E}| = 397$ .

In particular, any nontrivial cycle has period at least 1054. The first length not excluded by the present bound is 1054. That is a property of the bound, not evidence for a 1054-cycle.

*Proof.* The verified descent floor is a first-passage descent induction: every start  $2 \leq n \leq 10^6$  realizes a finite word with image strictly below the start, and strong induction on the image reaches 1. The longest first passage in the window has 253 steps (seed 78901); every iterate is an exact integer. Weisstein [5] records the same computational verification; the run here is a recomputation of that report.

The gap table computes  $o_{\min}(L)$  and  $n_{\max}(L)$  by exact integer arithmetic for every  $1 \leq L \leq 10^5$ . Corollary 4.5 at  $N_0 = 10^6$  excludes every  $L$  with  $n_{\max}(L) \leq 10^6$ . The complementary set in that range is exactly  $\mathcal{E}$ . The contiguous excluded

prefix is  $L \leq 1053$ . The record (one-sided best-approximation) lengths in range are

$$1, 3, 11, 19, 84, 569, 1054, 25781, 50508,$$

with  $n_{\max} = 3, 13, 52, 297, 5599, 58398, 1997197, 67410774, 420161535$ . The first six of those already satisfy  $n_{\max} \leq 10^6$  and are excluded. The first length not excluded by the present bound is  $L = 1054$ . The remaining elements of  $\mathcal{E}$  include multiples of 1054 and combinations of those one-sided best-approximation lengths; the defining algorithm and checksums are Appendix B.  $\square$

The former record lengths 84 and 569 are therefore not admissible at this floor. Membership of 1054 in  $\mathcal{E}$  means only that the present method does not exclude it.

## 5. Remarks

A start  $n \geq 2$  has a *descent certificate* if there exists a realized finite word  $w$  with  $J^{|w|}(n) < n$ . Even starts realize  $E$ :  $\lfloor \sqrt{n} \rfloor < n$ . An odd start with even image realizes  $OE$ , and  $J^2(n) \leq n^{3/4} < n$ . The starts not covered by those two words are exactly the odd-to-odd starts. If every start above 1 has some descent certificate, ordinary strong induction yields arrival at 1. That hypothesis is not proved.

The first odd-to-odd image expands, so ordinary strong induction cannot fire on the odd-to-odd class. A uniform run bound on expanding blocks is likewise unavailable: four consecutive expanding blocks occur already at

$$1999 \xrightarrow{OOE} 5169 \xrightarrow{OOOOEE} 50093 \xrightarrow{OOE} 193753 \xrightarrow{OOE} 887471,$$

and the relative slack of a single letter tends to 0 with the state (Section 2).

These results provide no control over the remaining lengths in  $\mathcal{E}$ , and in particular do not imply termination or exclude all cycles. It is open whether every start reaches 1, and open whether a cycle of an admissible leftover length exists.

Lean names for the theorems of Sections 2, 3, and 4 are in Appendix A. The formalization and the finance tables are pinned in Section 1.2. Theorem 4.6 is a verified computation.

## Appendix A. Lean names

The proofs in the text are ordinary integer arguments. The names below are the corresponding Lean theorems in `formal/Problems/Juggler/`, imported by `Problems.JugglerPaper`. A Lean identity is not a `native_decide` table; the tables of Section 3 are of the second kind. Theorem 4.6 is a verified computation outside Lean.

| Text                | Lean                                      |
|---------------------|-------------------------------------------|
| itinerary semantics | follows_iff_word,                         |
| Theorem 2.1         | image_eq_iterate, image_append            |
| Theorem 2.2         | image_monotone_of_follows                 |
| Corollary 2.3       | power_bound_word                          |
| Theorem 2.4         | power_bound_contracts                     |
| Theorem 2.5         | global_defect_identity                    |
|                     | global_defect_eq_zero_iff_localsTight,    |
|                     | global_defect_eq_zero_implies_monochrome, |
|                     | power_bound_eq_iff_extremal               |
| Theorem 2.6         | global_defect_append                      |
| Corollary 2.7       | image_eq_start_defectRatio                |
| per-step slack      | one_plus_eta_lt_succ_sq                   |
| Lemma 3.1           | odd_cell_unique                           |
| Theorem 3.2         | cycle_word_formally_expanding,            |
|                     | cycleMin_start_odd,                       |
|                     | cycleMax_start_even,                      |
|                     | cycleMin_not_end_odd,                     |
|                     | square_scale_superquadratic,              |
|                     | cycleMin_to_even_superquadratic           |
| Lemma 3.3           | lower_growth_word                         |
| Lemma 3.4           | oo_suffix_threshold,                      |
|                     | ooo_suffix_threshold,                     |
|                     | threshold_inherits_odd_append,            |
|                     | cycle_last_even_interval,                 |
|                     | no_cycle_odd_run_append_even              |
| Lemma 3.5           | no_cycle_word_oooeoe,                     |
|                     | no_cycle_word_ooooee                      |
| Theorem 3.6         | no_cycle_word_length_le_six               |
| Lemma 3.7           | no_cycle_word_ooooeoe,                    |
|                     | no_cycle_word_oooooee                     |
| Theorem 3.8         | no_cycle_word_length_le_seven,            |
|                     | with no_cycle_word_ooeooee,               |
|                     | no_cycle_word_oooeoee                     |
| Lemma 3.9           | cycle_trailing_evens_lt                   |
| Lemma 3.10          | lowerDenom_replicate_odd,                 |
|                     | odd_run_lower_growth                      |
| Lemma 3.11          | no_follows_seven_odds_of_lt256            |
| Theorem 3.12        | no_cycle_word_two_even_ee,                |
|                     | no_cycle_word_two_even_eoe                |
| Theorem 3.13        | no_cycleMin_gapped_three_even_ee,         |
|                     | no_cycleMin_gapped_three_even_eoe         |
| Theorem 3.14        | no_cycle_word_three_even_eee,             |
|                     | with no_cycle_word_oooooeeee              |
| Theorem 3.15        | no_cycle_word_three_even_eoe              |

| Text                                    | Lean                                                                       |
|-----------------------------------------|----------------------------------------------------------------------------|
| Theorem 3.16                            | no_cycle_word_three_even_eoeee                                             |
| Theorem 3.17                            | no_cycle_word_three_even_eooooe                                            |
| Theorem 3.18                            | no_cycle_word_three_even_eeoe                                              |
| Theorem 3.19                            | no_cycle_word_three_even_eoeoe                                             |
| Theorem 3.20                            | no_cycle_word_three_even_eooeee                                            |
| Theorem 3.21                            | no_cycle_word_gapped_three_even_ee,<br>no_cycle_word_gapped_three_even_eoe |
| Theorem 3.22                            | no_cycle_word_even_count_le_three                                          |
| Corollary 3.23                          | cycle_word_length_ge_eleven                                                |
| Lemma 3.21b                             | canonical run form; Theorem 3.2                                            |
| Lemma 3.21a                             | the case split of Theorem 3.22                                             |
| Lemma 4.1                               | log_le_two_log_add                                                         |
| Lemma 4.2                               | log_step_even, log_step_odd                                                |
| Lemma 4.3                               | cycleMin_log_envelope                                                      |
| Theorem 4.4                             | cycleMin_finance                                                           |
| short certificates (Section 5)          | even_finiteProgress,<br>odd_even_finiteProgress                            |
| no certificate $\Rightarrow$ odd-to-odd | no_finiteProgress_implies_odd_odd                                          |
| induction to 1                          | reachesOne_of_all_finiteProgress                                           |
| four-block chain                        | four_block_pe_1999                                                         |

## Appendix B. Admissible lengths

The record lengths of  $\gamma(L)$  through  $10^5$ , with the 6/5 bound  $n_{\max}$  of Section 4, are

| $L$   | $o_{\min}$ | $n_{\max}$ |
|-------|------------|------------|
| 1     | 1          | 3          |
| 3     | 2          | 13         |
| 11    | 7          | 52         |
| 19    | 12         | 297        |
| 84    | 53         | 5599       |
| 569   | 359        | 58398      |
| 1054  | 665        | 1997197    |
| 25781 | 16266      | 67410774   |
| 50508 | 31867      | 420161535  |

At the verified descent floor  $N_0 = 10^6$  the first six rows are excluded. The set  $\mathcal{E} = \mathcal{E}(10^6)$  is defined by Proposition 4.4a: for each  $1 \leq L \leq 10^5$ , compute  $o_{\min}(L)$  and  $n_{\max}(L)$  by exact integer arithmetic, and retain  $L$  if and only if  $n_{\max}(L) > 10^6$ . This produces 397 lengths. The first few are

1054, 2108, 3162, 4216, 5270,

and a typical combination is  $23757 = 22 \cdot 1054 + 569$ . The last is 99729. The complete list is the `lengths` array of the object with `"floor": 1000000` in `data/research/juggler/cycle_finance/exceptions.json`. The SHA-256 of that array, serialized as a JSON list of integers with no spaces, is `6d2c75fb6165f41123164122bd799e598c6ce0ba79d79d3af909cf400551f72c`. The SHA-256 of the whole file `exceptions.json` is `31f589e2353a26b13503d9fba603565fe3e6319030f6e429d8`. The SHA-256 of the first-passage file `floor.json` in the same directory is `5b1ce1eec61301cf5b4f969cd5b58954255194e5c7b21c08a518d71679af87fc`. Both files are regenerated by `python -m research.juggler_sequence.cycle_finance`. The finite leftover tables of Section 3 are the Lean `native_decide` evaluations named in Appendix A (`LeftoverEval.lean`, `LeftoverShort.lean`, `LeftoverFamilies.lean`).

## Appendix C. Composition of the global defect

**Theorem 2.6 (composition).** If  $u$  is realized at  $n$  and  $v$  is realized at  $m = J^{|u|}(n)$ , then  $\Delta_{uv}(n)$  is the sum of two power-gaps, one lifting  $\Delta_u(n)$  through  $3^{\#O(v)}$  and one lifting  $\Delta_v(m)$  through  $2^{|u|}$ . In particular  $\Delta_v(m)^{2^{|u|}} \leq \Delta_{uv}(n)$ , and every local remainder satisfies  $\rho_j^{2^j} \leq \Delta_w(n)$ .

*Proof.* Apply Theorem 2.4 to  $u$ , to  $v$ , and to  $uv$ , and expand the two power-gaps.  $\square$

Composition is polynomial, not additive. The Lean form is `global_defect_append`. The identity is not used by Theorem 4.4.

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## References

1. C. A. Pickover, *Computers and the Imagination: Visual Adventures Beyond the Edge*, St. Martin's Press, New York, 1991, ch. 40, p. 232.
2. C. A. Pickover, *The Mathematics of Oz: Mental Gymnastics from Beyond the Edge*, Cambridge University Press, Cambridge, 2002, ch. 45, pp. 102–106.
3. OEIS Foundation Inc., “Juggler sequence: if  $n$  even then  $\lfloor \sqrt{n} \rfloor$  else  $\lfloor n^{3/2} \rfloor$ ,” Sequence A094683 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A094683> (accessed 28 August 2026).
4. OEIS Foundation Inc., “Number of steps needed for  $n$  to reach 1 in the juggler sequence,” Sequence A007320 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A007320> (accessed 29 August 2026).
5. E. W. Weisstein, “Juggler Sequence,” *MathWorld*, <https://mathworld.wolfram.com/JugglerSequence.html> (accessed 29 August 2026).

6. OEIS Foundation Inc., “Largest value in trajectory of  $n$  under the juggler map of A094683,” Sequence A094716 in *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org/A094716> (accessed 29 August 2026).
7. V. Prasad and M. A. Prasad, “Estimates of the maximum excursion constant and stopping constant of juggler-like sequences,” ResearchGate preprint, 2025. <https://doi.org/10.13140/RG.2.2.14110.04168>.
8. J. C. Lagarias, “The  $3x + 1$  problem and its generalizations,” *Amer. Math. Monthly* 92 (1985), 3–23. doi:10.1080/00029890.1985.11971528.
9. J. C. Lagarias (ed.), *The Ultimate Challenge: The  $3x + 1$  Problem*, American Mathematical Society, Providence, RI, 2010.
10. R. E. Crandall, “On the “ $3x + 1$ ” problem,” *Math. Comp.* 32 (1978), 1281–1292. doi:10.1090/S0025-5718-1978-0480321-3.
11. K. R. Matthews and A. M. Watts, “A generalization of Hasse’s generalization of the Syracuse algorithm,” *Acta Arith.* 43 (1984), 167–175. doi:10.4064/aa-43-2-167-175.
12. J. L. Simons and B. M. M. de Weger, “Theoretical and computational bounds for  $m$ -cycles of the  $3n + 1$ -problem,” *Acta Arith.* 117 (2005), 51–70. doi:10.4064/aa117-1-3.
13. S. Eliahou, “The  $3x + 1$  problem: new lower bounds on nontrivial cycle lengths,” *Discrete Math.* 118 (1993), 45–56. doi:10.1016/0012-365X(93)90052-U.